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Privacy and personal data risk governance for generative artificial intelligence: A Chinese perspective

Xiongbiao Ye[a], Yuhong Yan[b,*], Jia Li[c,d], Bo Jiang[e]

[a] Law School, Central China Normal University, China
[b] School of Government, University of International Business and Economics, China
[c] School of Law, University of International Business and Economics, China
[d] School of International Studies, Hainan University, China
[e] School of International Law, China University of Political Science and Law, China

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ABSTRACT

The rapid development of generative artificial intelligence (AI) has attracted global attention and posed challenges to existing data governance frameworks. The increased technical complexity and expanded scale of data usage not only make it more difficult to regulate AI but also present challenges for the current legal system. This article, which takes ChatGPT's training data and working principles as a starting point, examines specific privacy risks, data leakage risks, and personal data risks posed by generative AI. It also analyzes the latest practices in privacy and personal data protection in China. This article finds that while China's governance on privacy and personal data protection takes a macro-micro integration approach and a private-and-public law integration approach, there are shortcomings in the legal system. Given that the current personal data protection system centered on individual control is unsuitable for the modes of data processing by generative AI, and that private law is insufficient in safeguarding data privacy, urgent institutional innovation is needed to achieve the objective of "trustworthy AI."

Following the advent of ChatGPT, OpenAI company introduced a new text-to-video generative artificial intelligence (AI) model named Sora in February 2024, once again capturing global attention. The introduction of Sora not only signifies advancements in the capabilities of generative AI for content creation but also intensifies global competition in the field of AI. In recent years, leading digital economies have accelerated the development of the AI industry, vying for global dominance. For instance, the United States (the US) published an updated plan for AI,[1] which focuses on critical areas within the AI industry and aims to maintain the US's leading position through national investments. The European Union (EU) released its strategic document for the AI industry in 2018,[2] highlighting the importance of relying on European values and inherent advantages to promote its AI industry. China has also prioritized generative AI as one of its key issue areas in the digital economy since 2023, leading to the launch of dozens of generative AI products by Chinese companies (iResearch, 2024). Other economies such as Japan, Germany, France, Russia, and South Korea are also formulating AI

* Corresponding author. No.10 Huixin East Street, Chaoyang District, Beijing, 100029, China.
E-mail address: 03388@uibe.edu.cn (Y. Yan).

[1] See The National Artificial Intelligence Research and Development Strategic Plan: 2019 Update. Available at https://www.nitrd.gov/pubs/National-AI-RD-Strategy-2019.pdf.

[2] See Communication from the Commission Artificial Intelligence for Europe. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=COM%3A2018%3A237%3AFIN.

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development strategies in recent years (Walch, 2018).

Guo et al. (2023) argued that generative AI poses multifaceted risks, including political, military, cultural, ethical, and data-related concerns, which have sparked widespread apprehension and increased attention to AI risk governance. Moreover, the ongoing global AI race extends beyond mere technological advancements, business strategies, and policy frameworks, it also encompasses a fierce debate over governance ideologies and regulatory approaches (Smuha, 2021). For instance, the US, aiming to regulate AI research, development, and utilization, issued the Executive Order - Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence in October 2023.[3] The EU's Artificial Intelligence Act, the world's first comprehensive regulations on AI, officially came into force on August 1, 2024. The Act aims to foster responsible AI development and to ensure that AI systems respect citizens' health, safety, and fundamental rights.[4] China took a significant step by enacting its first generative AI regulation in July 2023. The Cyberspace Administration of China (CAC),[5] the National Development and Reform Commission of the People's Republic of China (PRC), the Ministry of Science and Technology of the PRC, the Ministry of Public Security of the PRC, and two other ministries jointly published the departmental rules named the Interim Measures for Administration of Generative Artificial Intelligence Services (the AI Measures).[6] Furthermore, China's State Council revealed its plan to enact a specialized AI law in the 2024 Legislative Work Plan.[7] Inspired by it, the Chinese Academy of Social Sciences first published an unofficial but influential Model Law on AI 1.0 (Expert Proposal Draft).[8] Given the evolving nature of AI technology and the nascent state of AI regulatory practices, it remains crucial to review and compare different governance approaches adopted by various countries.

Among the various risks associated with generative AI, privacy and personal data risks stand out as the most immediate and serious challenges. On the one hand, ChatGPT has faced allegations of unauthorized personal data usage and data breaches. Victims of those lawsuits argue that such activities harm their privacy, personal data, and property rights.[9] On the other hand, data security risks also pose significant challenges to achieving trustworthy AI, which is crucial for the future development of the generative AI industry. Therefore, it's imperative to thoroughly analyze the privacy and personal data risks stemming from the development and use of AI. Additionally, urgent discussions are needed to address the shortcomings of existing privacy and data protection regulations in light of ongoing AI technology advancements. Notably, while this article discusses the privacy and personal data protection system from China's perspective, its findings and conclusions can serve as a reference for other countries exploring their regulatory strategies for generative AI.

1. Privacy and personal data risks in generative AI

1.1. Personal data for generative AI pre-training

In the digital economy era, the advancements of technology, as well as innovative business models, are closely intertwined with personal data. However, the widespread utilization of data has also led to a worldwide challenge concerning the protection of privacy and personal data.

Personal data refers to various types of information that can identify a natural person. Within the current Chinese legal system, there are two slightly different expressions for defining personal data. The Personal Information Protection Law (PIPL) defines it as "information related to identified or identifiable natural persons."[10] Meanwhile, the Civil Code of the People's Republic of China (CCPRC) describes personal data as information that can identify a natural person "by itself or in combination with other information."[11] This definition, influenced by the identifiability rule set by the EU's General Data Protection Regulation (GDPR), aligns with the universal concept of direct and indirect identifiers, despite variations in wording (Cao, 2022). Specifically, "identified" information directly pinpoints a particular individual, such as identification numbers or passport details. In contrast, "identifiable" information requires additional context or data to identify specific natural persons, such as behavioral patterns or internet browsing history.

OpenAI employs three primary types of data to train ChatGPT: data that is publicly available on the internet; data that it licenses from third parties, and data from its users or its human trainers.[12] Public information from the internet includes Wikipedia, books,

[3] See The Executive Order No. 14110, C.F.R. 75191. https://www.federalregister.gov/documents/2023/11/01/2023-24283/safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence.

[4] Artificial Intelligence Act (https://eur-lex.europa.eu/eli/reg/2024/1689/oj).

[5] The Cyberspace Administration of China (CAC) was established in May 2011 to enhance network regulation and address social issues arising from the rapid development of the internet. The primary functions of the CAC include implementing guidelines and policies for the dissemination of information online, promoting the legal framework for internet information dissemination, guiding, coordinating, and supervising relevant departments to strengthen internet content management, and investigating and penalizing illegal and unauthorized websites in accordance with the law (https://www.cac.gov.cn/2014-08/01/c_1111903999.htm).

[6] https://www.cac.gov.cn/2023-07/13/c_1690898327029107.htm.

[7] See State Council Legislative Work Plan for 2024. https://www.gov.cn/zhengce/zhengceku/202405/content_6950094.htm.

[8] See Model Law on Artificial Intelligence 1.0 (Expert Proposal Draft). https://dsj.hainan.gov.cn/zcfg/zybs/202308/t20230822_3479331.html.

[9] p.m. et al. v. OpenAI LP et al., (N.D. California 2023). https://assets.bwbx.io/documents/users/iqjWHBFdfxIU/rIZH4FXwShJE/v0.

[10] Personal Information Protection Law, Article 4. https://personalinformationprotectionlaw.com/PIPL/article-4.

[11] Civil Code of the People's Republic of China, Article 1034. https://english.www.gov.cn/archive/lawsregulations/202012/31/content_WS5fedad98c6d0f72576943005.html.

[12] OpenAI. *How ChatGPT and Our Language Models are Developed: Learn More about How We Develop Our Models and Apply Them in Products like ChatGPT*. https://help.openai.com/en/articles/7842364-how-chatgpt-and-our-language-models-are-developed.

journals, Common Crawl, Reddit links, and other sources, such as StackExchange and video subtitle datasets (Thompson, 2022). While this information does not explicitly confirm the presence of personal data in the pre-training datasets of generative AI, it does not mean that the training of generative AI products is irrelevant to personal data. The reasons are as follows:

First, the output of big language models includes personal data, and it indicates that developers have used personal data in the pre-training phase. For instance, Quach (2021) discovered that the output of GPT-2 included at least 0.1% of personal information or copyrighted material, including names, addresses, phone numbers, etc. Additionally, Sam Altman, CEO of OpenAI, acknowledged that some ChatGPT users could access other users' conversation history due to problems with the open-source database (Haughey, 2023). Second, generative AI enterprises collect and process users' data once their AI products are on the market. For example, ChatGPT's privacy policy states that it will collect users' data to "improve services and conduct research," "develop new programs and service," and "train the models that power ChatGPT."[13] Similarly, Ernie Bot, a generative AI product developed by the Chinese company Baidu, utilizes user inputs to enhance language understanding, recognition, and response capabilities.[14] Conversations with generative AI may not overtly involve direct identifiers like real names or phone numbers. However, they often touch upon users' life experiences, work status, recent thoughts, and interests—seemingly innocuous details that, when combined, can potentially reveal an individual's identity. Even traditionally considered "non-personal data" can be transformed into personal information through data analytics within legal frameworks (Supriyadi, 2023). In summary, these examples underscore the unavoidable utilization of personal data by generative AI, leading to privacy risks, personal data risks, and data leakage risks across different phases of AI project life cycles.

Research on personal data governance has often blurred the distinction between privacy and personal data protection. However, the two concepts should not be conflated for two reasons. First, countries typically regulate privacy and personal information protection separately through distinct legislation. For example, the EU's GDPR primarily addresses the processing of personal data, while privacy-related issues are within the jurisdiction of other laws, such as civil law (van Ooijen & Vrabec, 2019). Second, the legal attributes defining privacy and personal information protection differ. The EU, for example, elevates data protection to a constitutional level, asserting that "everyone has the right to the protection of personal data concerning him or her."[15] This right is considered a new fundamental human right, distinct from privacy in the aspects of concept, values, and means of protection (Tzanou, 2013).

In China, privacy is considered an absolute right, which resolves conflicts between rights holders and alleged infringers. However, personal data encompasses private, commercial, and public administrative values. Balancing these competing interests related to personal data cannot be fully managed within privacy regulations alone (Zhang, 2015). Consequently, most Chinese scholars recognize personal information as a distinct type of personality right, separate from privacy (Cheng, 2020). This perspective is officially reflected in Chinese laws. The CCPRC stipulates both the right to privacy as well as the rights and interests of personal information (Article 1032 and 1034, respectively). Also, it anticipates potential overlapping issues between privacy and personal data protection and stipulates that "the provisions on the right to privacy, or, in their absence, the provisions on the protection of personal information, shall be applied to the private personal information."[16] Taking into account the conflicts of interest, risk governance, and public values, the distinction between personal information and the right to privacy becomes even more pronounced, indicating that these two concepts should not be conflated.

1.2. Privacy risks in generative AI

In general, privacy encompasses an individual's right to seclusion. According to Chinese law, the object of privacy includes private spaces, private activities, and private information.[17] In the digital economy, the proliferation of big data, the Internet of Things (IoT), and AI technologies heavily rely on accessing and analyzing users' profiles to provide personalized services. However, such activities constantly expose users to electronic monitoring, infringing on their right to be free from public interference. Users may resist having their private thoughts and behaviors recorded because being recorded means risks of leak or misuse. In the case of Zhu vs. Baidu Inc.,[18] the plaintiff, Zhu argued that it was her private activity to visit web pages related to breast enlargement, abortion, and weight loss. The court of first instance held that an individual's online personal activities, including browsing history, constitute basic information about that person and are within the realm of personal privacy. Given that Baidu employed cookie technology to track the plaintiff's browsing history and to recommend relevant advertisements based on the plaintiff's browsing information, it undoubtedly constituted an infringement of the plaintiff's right to privacy.

As intelligent devices, such as self-driving cars, office gadgets, and home automation systems, become increasingly prevalent, they also encroach upon an individual's private moments of seclusion and personal space—whether in the bedroom, car, or office. For instance, when generative AI is employed for personalized house decoration, clothing design, diet planning, or health management, users "voluntarily" provide their data to the machine, or even allow it to observe and record their lives, residences, and behaviors. Similarly, in enterprise settings, customized AI software for employee management continuously collects work-related data, including work hours, status, and content. Beyond these traditional surveillance scenarios, generative AI models like Sora, ChatGPT, and Ernie

[13] See "How We Use Personal Information" Section of the OpenAI Privacy Policy. https://openai.com/policies/privacy-policy/.
[14] See Personal Information Protection Rules of Ernie Bot. https://yiyan.baidu.com/ER7FX3ATERRQT6.
[15] EU Charter of Fundamental Rights, Article 8. https://fra.europa.eu/en/eu-charter/article/8-protection-personal-data?page=1.
[16] Article 1034(3) of the CCPRC.
[17] Art. 1032 of the CCPRC.
[18] Zhu Ye v. Baidu Inc, Civil Judgment of First Instance, January 31, 2019, online (The Primary People's Court of Gulou District of Nanjing City, Jiangsu Province June 25, 2013) (China).

Bot delve even deeper—they monitor a user's thoughts. Conversations with generative AI, which reveal the innermost thoughts and emotions, contribute to the AI's training data set and shape its intellectual growth. Consequently, privacy risks associated with generative AI differ significantly from traditional surveillance, presenting more serious governance challenges for national regulators.

1.3. Data leakage risks in generative AI

Data leakage is a global concern. Despite enterprises' assurances of secure data management, inherent vulnerabilities in storage technology, management practices, and transmission processes occasionally result in breaches. When users engage with generative AI products and share their data, they assume the risk of leakage due to technical flaws or database attacks (Poremba, 2023), and other potential personal or property harm (Loos, 2019). For instance, in 2023, the US reported 15.4 million data fraud cases, leading to approximately $20 billion in losses (Buzzard, 2023). Similarly, China witnessed a 70% surge in telecommunication fraud cases in 2022, resulting in roughly CNY 35.37 billion in losses (Peng, 2021). Beyond financial crimes, data leakage offenses are evolving into more serious crimes, including human trafficking, kidnapping, and illegal detention.[19]

Generative AI significantly amplifies the risk of data leakage, directly exposing personal information to other users. Recent research reveals that when ChatGPT is repeatedly prompted to repeat the word "poem," it regurgitates entire passages from its training data. Additionally, it may inadvertently output people's names, contact details, and other private information from its training corpus.[20] In March 2023, due to a vulnerability in the open-source code it used, ChatGPT experienced a large-scale user data leakage incident. Although engineers promptly addressed the vulnerability, the incident triggered a crisis of confidence in ChatGPT's data security. The codes and models of generative AI rely on various open-source components, such as PyTorch, TensorFlow, and OpenSSL. If vulnerabilities exist in any of these components, security risks can propagate step by step, leading to systemic issues.[21] By June 2023, over 100,000 ChatGPT accounts had been compromised and sold on the dark web market,[22] posing significant threats to users' privacy and property.

1.4. Personal data risks in generative AI

Across the globe, countries recognize individuals' rights and interests in their data, including the right to be informed, the right to consent or object to processing, the right of access, the right of rectification, the right to erasure, etc. Recent policies and laws highlight that AI research, development, and applications must not infringe on personal data interests. For instance, Article 4(4) of the AI Measures mandates that generative AI services respect others' rights and avoid compromising privacy and personal data interests. However, the complexity and opacity of algorithmic mechanisms in machine learning pose challenges for explaining and assessing AI's data processing and protection measures (Ananny, 2015). Generative AI, a recent advancement in machine learning, presents similar personal data protection challenges. Generative AI consumes vast amounts of personal data while operating in a "black box" manner, where its internal workings and decision-making processes are opaque and not easily understood by users or even its creators. This opacity challenges data owners in exercising their data rights. Beyond the technical aspects, factors such as user-generated data during interactions with AI, multi-enterprise collaborative services, and unclear privacy protection policies contribute to the non-technical aspect of black box characteristics of generative AI (Pieters, 2011), posing extra threats to personal data protection.

Firstly, the right to be informed and the right to consent to process may be practically infringed upon in the context of generative AI. Lilian Edwards et al. (2024), after examining privacy policies of mainstream generative AI products, reveal that companies often neglect user privacy and personal information protection. Some generative AI products lack privacy agreements altogether, while others merely provide the parent company's general privacy policy, rather than specially tailored information for foundation models. This absence of clear privacy policies directly limits users' right to be informed. Even if a user agreement outlines how the generative AI service provider accesses and uses personal information, understanding such policies can be challenging without knowledge of AI, machine learning, cybersecurity, and legal nuances. Instructions related to technologies like cookies can be particularly perplexing for average users.[23] Once individuals sign up for a generative AI product, their data is handed over to the controller or developer. Losing control over their data exposes them to unforeseen risks. Moreover, generative AI heavily relies on personal data for research and development, which individuals do not voluntarily provide. In essence, the infringement on individuals' right to be informed and the right to consent to process occurs as early as the research phase, leaving users unaware of its implications.

Secondly, the right of access and the rights of rectification and completion are highly limited. The collector provides users with information query services, allowing them to check how their data is recorded and utilized. However, users can neither access the history of their conversations with the AI nor understand how this data is stored and processed. Instead, they can only access basic data,

[19] See the Supreme People's Procuratorate of the People's Republic of China (2023). *检察机关打击治理电信网络诈骗及其关联犯罪工作情况 [Prosecutorial Efforts in Combating Telecommunication Network Fraud and Related Crimes]*. https://www.spp.gov.cn/xwfbh/wsfbt/202311/t20231130_635181.shtml#2.

[20] See Tiernan Ray (2023). *ChatGPT Can Leak Training Data, Violate Privacy, Says Google's DeepMind*. https://www.zdnet.com/article/chatgpt-can-leak-source-data-violate-privacy-says-googles-deepmind/.

[21] See *All about ChatGPT's First Data Breach, and How It Happened*. https://www.pluralsight.com/blog/security-professional/chatgpt-data-breach.

[22] See *Group-IB Discovers 100K + Compromised ChatGPT Accounts on Dark Web Marketplaces; Asia-Pacific Region Tops the List*. https://www.group-ib.com/media-center/press-releases/stealers-chatgpt-credentials/.

[23] *Supra* note 14.

such as their account names, phone numbers, and login locations, which is far less than the information acquired by the collector (Bufalieri et al., 2020). Worse still, as generative AI models have transitioned from supervised learning to unsupervised learning, gaining the ability to autonomously process data and learn independently, the collector is often unaware of what data generative AI models use, how they process data, and what outputs they generate (van Genderen, 2017). Consequently, users cannot effectively exercise their right to access and are unlikely to be able to correct or complete their data.

Thirdly, the realization of the right to erasure faces significant challenges. Large language models exhibit remarkable autonomous learning capabilities. When a user inputs information, it becomes raw material for the model's training and growth. This process, however, raises questions about whether a user's input information can be effectively separated and deleted from the training corpus upon the user's request. Current research indicates that the process of data deletion or "forgetting" within AI systems is technically intricate and may be nearly impossible to achieve (Villaronga et al., 2018). Moreover, the scope of personal data, based on the criteria for "personally identifiable information" remains ambiguous. Interactive data generated during users' interactions with generative AI results from a collaboration between the user and AI platforms. Even though the interactive data can be identified by the AI platform, it is questionable whether users have a unilateral right to request the erasure of their data. In summary, given that generative AI functions as a black box that combines intricate algorithms with copious amounts of data, it indeed poses substantial risks to an individual's right to personal data.

2. Risk governance system for the protection of privacy and personal data in China: from a macro- and micro-level perspective

In July 2023, the AI Measures, which specifically addresses the privacy and personal data risks associated with generative AI, was introduced to guide and regulate the development of the generative AI industry in China. The AI Measures requires high-quality training data and mandates the protection of data subjects' rights and interests during the development of generative AI. In addition, this regulation stipulates that all processing activities involving trained data for AI must comply with existing Chinese legislation, including PIPL, the Cybersecurity Law (CSL),[24] the Data Security Law (DSL),[25] and other norms related to personal data.[26] On the one hand, the AI Measures represents specialized legislation tailored to address the unique challenges posed by generative AI. On the other hand, it could be characterized as emergency legislation introduced just eight months after ChatGPT's release. Chinese legislators, lacking a precise understanding of fundamental theories, professional technologies, application scenarios, and potential risks associated with generative AI, have framed the AI Measures as guiding principles rather than detailed rules. These provisions create a legal framework to facilitate AI industry development. As regards data processing, service provider obligations, user rights, regulatory mechanisms, and legal liabilities related to generative AI, the AI Measures does not prescribe specific rules. Instead, it stipulates that these matters should be addressed in conjunction with existing laws, such as the PIPL. In other words, the AI Measures serves as a bridge connecting generative AI with the established legal system for privacy and personal data protection, effectively integrating laws like the PIPL and the DSL into the realm of generative AI governance.

Section 2 analyzes such a legal system in the era of AI from both the macro-level and micro-level perspectives. At the macro level, China's policies and theoretical development, governance ideas, legal system of privacy, and personal data protection will be analyzed to contribute to a comprehensive understanding of China's legal framework concerning generative AI. On the micro level, the focus narrows to specific rules. This article explores how Chinese law conceptualizes personal data, safeguards the rights and interests of privacy or personal data, and establishes regulatory measures. These specific rules serve as the foundation for further assessments of any inadequacies in the existing legal framework.

2.1. Macro framework for privacy and personal data protection: collaboration of general legislation and specific legislation

Over a prolonged period, China's underdeveloped digital industry resulted in fragmented and incoherent legislation on privacy and personal data protection, making foreign scholars concerned about its future legislative direction in this field (Greenleaf, 2012). In the past few years, however, the widespread adoption of the internet, the digital transformation of China's economy, and the emergence of novel social problems related to data such as personal data leakage, data-driven price discrimination (Bian, 2024), and unfair competition among internet companies (Hou, 2018), have compelled China to recognize the importance of cyber governance. Consequently, China has adjusted its national development strategy from "focusing solely on economic and technological development while neglecting cyber order and data security" to "focusing on both economic growth and social governance."[27] In addition, intensified global competition and growing cybersecurity threats have also driven China to progressively strengthen its legislation for network and personal data protection since the 2010s.

On April 15, 2014, President Xi Jinping of China introduced "a holistic view of national security" during a meeting of the National Security Commission for the first time,[28] which requires systematically integrating traditional security across various domains,

[24] See Cybersecurity Law of the People's Republic of China. https://www.gov.cn/xinwen/2016-11/07/content_5129723.htm.
[25] See Data Security Law of the People's Republic of China. http://en.npc.gov.cn.cdurl.cn/2021-06/10/c_689311.htm.
[26] Art. 7 of The AI Measures.
[27] See the 14th Five-Year Plan for the Development of the Digital Economy. https://www.gov.cn/zhengce/content/2022-01/12/content_5667817.htm.
[28] See A Holistic View of National Security. http://en.qstheory.cn/2020-12/04/c_978625.htm.

including politics, military, homeland, economy, culture, society, science and technology, cyberspace, AI, and data. This holistic approach is an inclusive system, emphasizing both traditional and non-traditional issues (Wang, 2024). To implement this approach, China has accelerated legislative activities in the security domains. A significant milestone occurred with the enactment of the National Security Law (NSL) in 2015. While not the first legal document involving data security, the NSL incorporates China's latest security notions and establishes innovative theories for shaping a new security legal system in the modern era, showing its attribute of a basic law (Yang, 2019). Particularly, Article 25 of the NSL mandates that "the state shall build a network and information security guarantee system, improve network and information security protection capability,"[29] serving as a source of law for subsequent legislation in the field of network and information security. After the NSL, more laws and regulations have been published to enhance this legal framework. Significantly, the National People's Congress and the Standing Committee of the National People's Congress, the highest legislative bodies in China, have enacted several crucial laws in the field of data security, including the CSL in 2016, the CCPRC in 2020, the DSL in 2021, and the PIPL in 2021.

Although China's legislation concerning digital data exhibits overlaps, different laws serve distinct objectives and regulatory priorities. These laws—guided by the holistic national security approach—reflect political position, social culture, legal traditions, and an understanding of data's legal attributes. Structurally, they follow a hierarchical layout of "cybersecurity - data security - personal data security - personal data security in the field of generative AI," forming a framework that progresses "from the whole to parts" and "from general fields to specific ones." Specifically, the CSL focuses on overall network security, delineating requirements for network infrastructure, operational standards, and data transmission. The DSL, zooming in on the data life cycle, regulates security matters related to data acquisition, transmission, storage, processing, and destruction. The PIPL, bearing a strong resemblance to the EU's GDPR, introduces new rights and interests related to personal data. It establishes detailed rules for users' rights, data processors' obligations, legal regulations, and responsibilities. Overall, China's theoretical and legislative system for data governance and personal data protection resembles a "pyramid," as in Fig. 1, with the CSL, the DSL, and PIPL serving as three essential "pillars" applicable to all digital economic activities within the country. Notably, the concepts and legal norms at the lower levels of the pyramid are not less significant or less effective but rather encompass a broader range of areas. Following a comparative analysis of China's PIPL and the EU's GDPR, Li and Chen (2024) contend that, given China's distinctive cultural, political, and legal traditions, the country has, after witnessing extensive theoretical and practical development, adopted a unique approach to personal information governance. Specifically, within China's digital governance framework, the prioritization of privacy and personal information protection is subordinate to national security and national data sovereignty.

China's data governance policies and legal system are intricately woven into the fabric of national security, while also considering broader societal needs such as social development and economic progress. Key milestones in this process include the issuance of the New Generation Artificial Intelligence Development Plan (the Plan) by the Chinese government in 2017.[30] The Plan outlines critical objectives, including the establishment of AI science and technology innovation systems, fostering a smart economy, and creating a safe and convenient intelligent society. It also emphasizes the need for laws, regulations, and ethical norms that promote responsible AI development, alongside the implementation of an AI security supervision and evaluation system. In 2019, the National New Generation Artificial Intelligence Governance Specialist Committee published the Governance Principles for a New Generation of AI: Develop Responsible AI (the Governance Principles), which clarifies the objectives of AI governance, such as promoting the healthy development of a new generation of AI, better coordinating the relationship between development and governance, ensuring that AI is safe/secure, reliable, and controllable, and promoting economically, socially, and ecologically sustainable development.[31] Additionally, the current legal framework prioritizes safety, while also taking into account various social interests. It aims to coordinate the diverse interests of all parties through reasonable rule arrangements. A central contradiction arises between the interests of enterprises and individuals. For instance, how can data property rights be designed to protect enterprises while respecting individual rights? How to address the rigidity of the "informed consent" rule that impacts the full utilization of personal data by enterprises? Within the field of generative AI governance, how to disclose the rules for protecting and utilizing personal data has caused controversy. We will delve into this issue later in the following section.

In addition to constructing a data security and personal data protection framework that addresses both macro and micro aspects of data governance, China acknowledges the multifaceted risks posed by emerging technologies and evolving business models. Given the inherent challenges for legislators to predict the precise trajectory of technological development, the modes of personal data processing, and the potential social issues that may arise, China adopted a specialized gap-filling legislative strategy to address specific risks in the digital era. This approach diverges from the traditional future-proofing method, which relies on vague terms or general provisions to enhance interpretability (Ranchordas & van 't Schip, 2020). The choice to adopt this strategy stems from two key reasons. First, relying on vague or principled provisions may result in the cognitive decline of individuals and enterprises on their legal obligations. For instance, since the definition and scope of "personal data" remain subjects of ongoing global discussion, enterprises are struggling to determine the lawfulness of their data processing activities. Second, China is a civil law country where the People's Court

[29] See National Security Law of the People's Republic of China. https://www.gov.cn/zhengce/2015-07/01/content_2893902.htm.

[30] See Notice of the State Council Issuing the New Generation Artificial Intelligence Development Plan. https://www.gov.cn/zhengce/content/2017-07/20/content_5211996.htm.

[31] See Governance Principles for a New Generation of Artificial Intelligence: Develop Responsible Artificial Intelligence. https://www.most.gov.cn/kjbgz/201906/t20190617_147107.html.

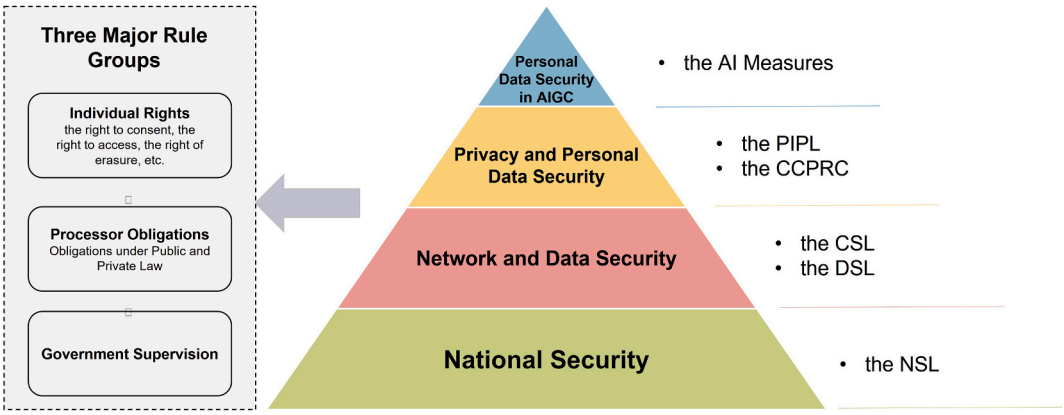

Fig. 1. The rule system and governance framework for privacy and personal data protection in China.

does not create laws, but only applies and interprets them.[32] The absence of clear and specific legal rules undoubtedly restricts judges' performance on legal applications.

The AI Measures is "gap-filling" legislation. Without a bill governing the AI industry, China promptly introduced the AI Measures to tackle privacy and personal data risks arising from generative AI development. Article 4 of the AI Measures emphasizes that generative AI service providers shall respect privacy and safeguard individuals' rights and interests regarding their data. Article 7 stipulates that consent from individuals must be obtained when the training data for generative AI involves personal data unless otherwise provided for by other relevant laws and regulations. Article 11 prohibits generative AI service providers from collecting non-essential personal data, unlawfully retaining user inputs and usage records, or sharing users' data with third parties without authorization. Article 11 also reaffirms that users have the right to access and make copies of their data, and request the processor to correct, complete, or delete their data during the utilization of generative AI. Moreover, the AI Measures takes precedence in governing personal data within the generative AI context as the specialized legislation with complementary general provisions under different laws such as the PIPL.

2.2. Micro framework for privacy and personal data protection

2.2.1. Synergy between public law and private law

Concerning the legal attributes of personal data interests, there are two different points in the Chinese academic community. The first perspective considers personal data interests as a new form of civil right, encompassing both personality and property interests (Wang, 2012). The second perspective views personal data interests as a constitutional right safeguarding the freedom and dignity of citizens (Zhao, 2019). These different perspectives correspond to two distinct modes of personal data protection. The first mode considers user "consent" as an expression of intent within civil law, asserting that individuals can exercise complete control over the flow of their data (Fuchs, 2011). Consequently, individuals shall be responsible for their privacy and data security. In contrast, the second mode recognizes that data risks pose a threat not only to individuals, but also to society, and thus, the state shall be responsible for the protection of personal data (Lynskey, 2014).

These perspectives on the nature of the right to personal data and data governance modes have significantly influenced China's personal data legislation. The first view of personal data protection is reflected in the provisions "Personal Information Interests Protection" within the CCPRC and Chapter 4 "Individual Rights in Personal Information Processing Activities" of the PIPL. Meanwhile, the second perspective aligns with Chapter 5 "Obligations of Personal Information Processors" of the PIPL and Chapter 4 "Obligations of Data Security" of the DSL.

Even though each perspective on personal data protection interprets the purpose, nature, and role of relevant rules in both the PIPL and the DSL differently, the theoretical debates have not significantly affected the practical application of the enactments. China's personal data protection system demonstrates a harmonious interplay between public and private law. While an individual claims his or her right to personal data mainly through the private law approach, the processor must fulfill its obligations of personal data protection under public law norms. In other words, the two facets—individual rights and the data processor's responsibilities—are essentially two sides of the same coin, operating in tandem without conflict.

2.2.2. A framework comprised of individual rights, processor obligations, and government regulation

Within the legal framework of synergy between public law and private law, China's privacy and personal data protection system can be categorized into three major rule groups, namely, individual rights, processor obligations, and government supervision (See Fig. 1).

[32] See Provisions of the Supreme People's Court on Citation of Such Normative Legal Documents as Laws and Regulations in the Judgments. https://www.court.gov.cn/fabu/xiangqing/73.html.

The rule group "individual rights" borrows relevant provisions from the EU's GDPR, and refers to the theory of personality rights and declaration of assent within the civil law of China. It prescribes the right to personal data, including the right to consent, the right of access, the right of correction and erasure, etc.

Corresponding to the first rule group, the second rule group "processor obligations" outlines the following obligations: (1) to obtain user consent; (2) to provide detailed explanations regarding the scope of data acquisition, and the purpose, methods of data processing; (3) to cooperate with users in inquiring, copying, or deleting their personal data; and (4) to safeguard personal data properly to prevent data leaks. In addition, given that safeguarding the security of personal data intertwines with national security and public interests, the processor is required not only to cooperate with data subjects to enjoy their rights, but also to fulfill public law obligations: (1) to encrypt and de-identify personal data and adopt categorized management based on risks and impacts to individual rights and interests[33]; (2) to appoint personal information protection officers to supervise personal information handling activities[34]; (3) to evaluate, record, and engage in audits of their data processing practices[35]; (4) to disclose relevant information and accept external supervision.[36]

To ensure processors' compliance with their obligations and maintain data security, the third rule group "government supervision" establishes a personal data supervisory mechanism centered on the CAC. It clarifies the supervisory duties and responsibilities of the supervisory departments and prescribes legal liabilities for violations of the PIPL. It's worth noting that, due to the widespread application of generative AI products, Article 16 of the AI Measures mandates that supervisory departments in cyberspace administration, development and reform, education, science and technology, industry and information technology, public security, radio and television administration, and press and publication administration regulate generative AI services under their respective duties, forming a collaborative regulatory approach.

3. Shortcomings in China's privacy and personal data protection system

Although China has established a legal framework for privacy and personal data protection, the existing regulatory concepts, governance modes, and specific rules have fallen behind the evolving demands posed by generative AI from both public and private law perspectives. As a result, this legal system neither effectively addresses privacy and personal data risks associated with generative AI, nor provides comprehensive support for the protection of personal data interests and other related rights.

3.1. Shortcomings in privacy protection

The Chinese law distinguishes privacy interests and personal data interests, addressing data privacy challenges arising from digital technologies through the privacy regime in civil law, and taking other personal data issues under the personal data protection regime. This institutional design, however, fails to sufficiently address the intrusion of generative AI into data privacy. The history of privacy suggests two meanings of this concept: freedom from intrusion and active control. The former emphasizes the inaccessibility of privacy, while the latter emphasizes the freedom of individuals to make decisions related to friendship, sexuality, marriage, procreation, religion, and other privacy matters (Allen, 1999). China's privacy theory often takes a standpoint of private interests to interpret privacy as a specific personality right under civil law (Zhu, 1997), and rarely delves into the deeper relationship between privacy and fundamental human rights, such as personal freedom and human dignity. Moreover, privacy guaranteed by Article 1032 of the CCPRC refers to negative privacy, granting individuals the right not to be disturbed. In other words, only private activities, spaces, and information unknown to others are protected by Chinese law. Such privacy rules are undeniably outdated in the era of AI and big data for the following reasons.

First, the utilization of generative AI poses challenges to existing rules to protect individuals' negative privacy regarding their personal information. Negative privacy protects personal information unknown to others, requiring a subjective intent of data subjects to keep it confidential. However, various AI applications prompt users to voluntarily share private data. For instance, customer relationship management systems integrated with generative AI technologies can leverage historical data to forecast future trends, thereby assisting organizations in tracking customers, proactively delivering services, and formulating sales strategies. However, this also necessitates that customers continuously supply a diverse array of data, encompassing personal experiences, ideas, and needs. Browsers with content generation functions may inadvertently expose private information through search keywords (e.g., property division, child support, or divorce agreement). AI service providers explicitly state in the user agreement that personal data will be used for research and development. By granting AI access to private information, the users effectively relinquish their right to secrecy. Furthermore, when Chinese courts determine a privacy infringement, the infringement needs to reach the level of "disturbance of the peace of life," requiring privacy violations to significantly impact an individual's peace of mind. For instance, when one person's dissemination of another private photographs and indecent videos adversely impacted the academic and personal life of the person, this action was deemed a violation of the right to privacy.[37] This underscores the challenge of determining privacy breaches in the era

[33] Article 51 of the PIPL (https://personalinformationprotectionlaw.com/PIPL/article-51).
[34] Article 52 of the PIPL (https://personalinformationprotectionlaw.com/PIPL/article-52).
[35] Article 54 of the PIPL (https://personalinformationprotectionlaw.com/PIPL/article-54).
[36] Article 55 of the PIPL (https://personalinformationprotectionlaw.com/PIPL/article-58).
[37] Zhu v. Tang, A Dispute over Privacy Right, Civil Judgment of First Instance, January 18, 2024, online (The Primary People's Court of Fengcheng City of Liaoning Province June 27, 2023) (China).

of big data.

Second, the utilization and processing of public information by generative AI present challenges to existing rules in safeguarding individuals' rights and interests related to publicly available personal data. A significant portion of generative AI training data originates from publicly available internet data, including information voluntarily shared by individuals (e.g., photographs and videos posted by users), and data disclosed by the government (e.g., business registration details, credit information, and judicial documents). Given the substantial personal information contained in these datasets, they closely intersect with individuals' right to privacy. In China, a predominant theory holds that even if individuals possess certain interests in their publicly available data, the exercise of these interests must be subordinate to the public's right to be informed, freedom of expression, and considerations of public interest. Under current Chinese law, Article 1032 of the CCPRC requires private information to meet both "privacy" and "unwillingness to be known by others" criteria for privacy protection. However, the disclosure of information – regardless of whether it's actively posting one's life, thoughts, and photographs on social media, or neutrally revealing personal information in publicly available judgments – disqualifies such information from the abovementioned criteria. In addition, Article 27 of the PIPL, which aims to balance privacy and public interests, specifically addresses the disclosure of personal information under which processors are endowed to obtain and process personal information without individual consent within a reasonable scope. Disclosure weakens privacy and personal control but strengthens public access (Liu, 2022), making such data even flow into the public domain (Anderson, 2011) and becoming freely accessible to individuals, enterprises, and organizations. For instance, the State may decide to disclose certain legal judgments based on public policy considerations, such as enhancing social oversight and disseminating legal knowledge. Consequently, these documents, including the personal information they contain, will enter the public domain, where they can be utilized by societal actors within reasonable limits (Zhang & Chang, 2022). Given that data is now recognized as a new factor of production,[38] it partly explains why China's data governance policies are evolving from unidimensional protection of individual interests to emphasizing both individual rights and public values.

However, these provisions also raise significant concerns. To start with, the term "public" lacks a clear definition, causing uncertainties in balancing between privacy protection and data utilization. Take WeChat, one of the most popular instant messaging software in China, as an example. When WeChat users choose to share personal information (such as selfie photos) exclusively with specific friends through the Moment feature,[39] they may argue that they intentionally limit the audience for their data. In other words, they don't mean to relinquish their right to privacy. However, other market players, in particular generative AI developers, may take an objective standard for publicly available data under Article 27 of the PIPL (Ding, 2024). Moreover, generative AI's training corpus often includes data from the internet, and individuals are usually unaware of how their publicly available data is processed, making it difficult to exercise their right of erasure.

3.2. Unclear accountability mechanisms for data leakage

China's current data legislations, including the CSL, the DSL, and the PIPL, aim to guide data processors in establishing robust data storage security systems and minimizing data leaks. For example, Article 21 of the DSL mandates the establishment of a categorized and graded protection system for data, based on the importance of data in economic and social development, and its impact on national security, public interests, and individual rights. However, since the DSL addresses data security on the internet sector and offline world, it does not stipulate in detail the criteria and methods of data categorization, leaving regulatory space for other laws and technical standards (Feng, 2024). As for personal data, Article 51 of the PIPL sets more explicit rules by requiring processors to manage personal information by classification, according to the purpose and method of processing, type of personal information, impact on individual rights and interests, possible security risks, etc. Article 57 of the PIPL establishes risk warning and reporting mechanisms to proactively address data leakage and unauthorized processing risks.[40] Also, Articles 33 and 34 of the Model Law on AI 1.0 (Expert Proposal Draft) introduce safety and remedial notification obligations for AI developers and providers and emphasize risk monitoring and remediation for security defects and vulnerabilities.

The aforesaid rules offer guidelines for generative AI developers and providers to process and store data. However, illegal data processing or data leakage that happens in countries lacking mature market competition mechanisms rarely has an immediate impact on consumer choices, such as forcing users to abandon the app that illegally processes their data. Given that, a network special campaign is frequently launched in China by regulatory departments to combat data breaches. For instance, several departments jointly issued in 2019 a Notice on the Special Campaign Against Illegal Collection and Use of Personal Information by Apps to enhance accountability and law enforcement. [41]

Notably, ChatGPT has been reported to be at risk of leaking users' data (Mudaliar, 2024). Severe data leakage incidents have also

[38] See Opinions of the CPC Central Committee and the State Council on Improving the Systems and Mechanisms for Market-based Allocation of Factors of Production. https://www.gov.cn/zhengce/2020-04/09/content_5500622.htm.

[39] Moment is a feature within the chat software WeChat. Users can post text, pictures, and videos. They have the option to set the visibility permissions to specific people or all friends. Additionally, users can specify the display duration for the relevant information, ranging from three days to all time.

[40] This provision mandates immediate remedial measures and notification when personal information is leaked, falsified, or lost. Refer to https://personalinformationprotectionlaw.com/PIPL/article-57.

[41] See Notice on the Special Campaign against Illegal Collection and Use of Personal Information by Apps. https://www.cac.gov.cn/2019-11/07/c_1574658334765452.htm.

occurred in China. Addressing the social risks caused by data leakage and providing relief to affected users are critical issues that require resolution. At present, Chinese law mainly stipulates administrative liability for data leakage accidents. Articles 44 and 45 of the DSL stipulate that when relevant competent authorities identify major security risks in data processing activities, they may conduct interviews with relevant organizations and individuals, requiring them to take corrective measures to eliminate hazards. Those who refuse to comply or cause significant data leaks or other serious consequences may face fines ranging from CNY 500,000 to CNY 2 million, along with potential additional penalties. The PIPL, designed specifically for personal information governance, imposes stricter penalties on violators. Article 66 of the PIPL allows relevant competent authorities to order corrections, confiscate unlawful income, and impose fines of up to CNY 50 million or 5% of the violator's last year's annual revenue if personal information is processed in grave violation of the law.

However, due to the complexity of social impact, the harm to the rights and interests associated with data leakage, and the difficulties of determining the ever-evolving forms and applications of technologies, these rules, along with other supporting rules, fail to clearly define the term "grave violation,". To date, China's legislators and People's Courts have developed four criteria for determining grave violations. First, the amount of leaked data. Given that a greater amount of leaked data implies higher risks, providing personal data of over 1 million people abroad may trigger a data transfer security assessment.[42] Second, the types of personal data. Given that sensitive data and important data are of greater importance, a small-scale leakage of such data may be considered serious. Third, the social consequences of data leakage, particularly its impact on national security and public safety.[43] Fourth, the intent of processors, that is, whether they act intentionally or negligently. Given China's supportive policies for the generative AI industry, it remains uncertain whether local generative AI enterprises will face heavy fines for data leakage. The Model Law on Artificial Intelligence 1.0 (Expert Proposal Draft) suggests that China aims to lower penalty standards for data leakage in the AI field, adjusting general violation penalties to less than CNY 1 million and grave violation penalties to less than CNY 50 million or a fine of less than four percent of the previous year's turnover.[44]

Regarding civil liability, although Article 69 of the PIPL allows users to resort to justice to protect their data rights and interests, users face theoretical challenges when bringing such litigation, making it difficult for them to obtain effective relief. The primary difficulties lie in determining the extent of damage resulting from personal data infringement. According to Article 69 of the PIPL, individuals can seek compensation for all the losses suffered from unlawful processing activities, including material and non-material damages, similar to the personal data infringement damage stipulated in Article 82(1) of the GDPR. The theory of full compensation in tort law suggests that damages should be the difference in the victim's interests before and after the infringement. However, it's challenging to precisely assess this difference. The EU faces similar challenges in interpreting Article 82(1) of the GDPR since there is no clear guidance on determining non-material damage therein. One possible interpretation, inspired by the damage to personality rights, is to treat non-material damage as "pain and suffering and loss of amenities." Nonetheless, this flexible interpretation cannot fully capture the nature of personal data infringement (Knetsch, 2022).

In China, personal data infringement litigation often arises from data leakage accidents, and the victims typically claim two types of losses: (1) mental distress resulting from harassment due to phone number leakage, and (2) financial losses due to fraud resulting from data leakage. For the former situation, Article 1183(1) of the CCPRC allows the victim of the tort to seek compensation for serious mental distress.[45] However, the mental distress caused by leaked phone numbers often falls short of severely affecting an individual's peace of life and causing significant mental distress. In the latter situation, while fraudulent losses may be recognized by some courts as part of personal data leakage damage, the victim at fault may still be found partly responsible for the losses.[46] Chinese scholars tend to adopt an abstract risk-based approach to identifying personal data leakage damage, under which data exposure is considered a danger (Loos, 2019). Following this approach, damage arising from data leakage includes an increasing risk of data leaks, expenditures on risk prevention, and risk-induced anxiety. However, the lack of clear rules to determine personal data leakage damage in the CCPRC indicates that the Chinese courts will grapple with this issue for the foreseeable future.

Besides, existing laws related to generative AI fall short of effectively addressing the subject of responsibility. Since generative AI industries involve various stakeholders, including researchers, developers, and providers, this issue has attracted more attention worldwide. Article 21 of the AI Measures states that if (generative AI) service providers violate the provisions of the Measures, they shall be penalized by the competent authorities under the relevant laws. However, this provision falls short in several aspects. To start with, since data leakage can occur through different subjects at different stages, it's unfair to solely require generative AI service providers to take responsibility. Second, under circumstances where multiple enterprises jointly provide generative AI services, this provision does not make it clear how to distribute the liabilities. Third, this provision does not specify the responsibility of the executives or other persons in charge, thus other laws such as Company Law or Criminal Law of the PRC may be applied to determine

[42] Measures for the Security Assessment of Outbound Data Transfer, Article 4(2).

[43] For instance, when DiDi company, a popular Chinese taxi app, sought to list on the New York Stock Exchange, Chinese authorities suspected the cross-border transfer of its data might jeopardize national security, and took a joint investigation. Based on the investigation results, DiDi was fined approximately CNY¥ 8 billion and was ordered to delist from the US. See Decision of the CAC on Administrative Penalties Related to Cybersecurity Review of DiDi Global Corporation. https://www.cac.gov.cn/2022-07/21/c_1660021534306352.htm.

[44] *Supra* note 8.

[45] Liu Zhiqing v. Shanghai Qi Yu Information Technology Co., Ltd., A Dispute Over Personality Rights, Civil Judgment of Second Instance, April 26, 2022, online (The Intermediate People's Court of Xiamen City of Fujian Province January 20, 2022) (China).

[46] Zhou Yuchan v. Guangdong KK E-commerce Co., Ltd., A Dispute over internet Tort Liability, Civil Judgment of Second Instance, June 20, 2019, online (The Intermediate People's Court of Shenzhen City of Guangdong Province May 10, 2019) (China).

their responsibility.

Existing laws and regulations such as the PIPL, the Provisions on Administration of Algorithmic Recommendation in Internet Information Services 2021, and other legal norms also fail to adequately address the responsibility for personal data leakage. Article 66 of the PIPL outlines penalties for unlawful data processing, including rectification orders, warnings, confiscation of illegal gains, and fines. However, these penalties primarily target enterprises associated with relevant network products or services. The challenge lies in the complex affiliations of large internet companies, where subsidiaries provide products while ultimate business decisions rest with the parent company. The Provisions on Administration of Algorithmic Recommendation in Internet Information Services 2021encounter similar challenges.[47] Article 31 of this regulation does not specify the responsible subject for data leakage. It merely states that if an algorithmic recommendation service provider violates relevant regulations, it shall be dealt with under the laws and administrative regulations. However, there is no specific reference point within existing law. Chinese scholars, having recognized the distinction between AI developers and service providers, have separately defined the obligations and responsibilities of each group. For instance, the Model Law on AI 1.0 (Expert Proposal Draft) suggests that if an AI model infringes on personal rights and interests and causes damage, the researcher, developer, or provider of it shall bear tort liability respectively, unless it can be proven that it is not at fault.

3.3. Inadequate code of conduct for data processing activities

Large language models are opaque black boxes, and even their developers cannot fully explain how the models process data. Also, although the cooperation between the individual rights rule group and the processor obligations rule group tentatively provides a framework to regulate the data processing activities of generative AI service providers in China, the existing rules are abstract and principled, falling short of fully addressing the personal data risks posed by generative AI.

Firstly, practical challenges arise when applying the minimization principle in the field of generative AI. For instance, given that data collection and processing activities inherently pose risks to users, Article 41(2) of the CSL prohibits network operators from gathering personal data unrelated to the services they provide. Similarly, Article 6(2) of the PIPL also emphasizes that the collection of personal data should be limited to the smallest scope for realizing the purpose, and prohibits excessive personal data collection. To effectively implement the principle of data minimization, four departments including the CAC collaboratively issued the Provisions on the Scope of Necessary Personal Information for Common Types of Mobile Internet Applications in 2021 (the Provisions 2021), which outlines the necessary personal data requirements for 39 common types of apps in Article 5. For instance, this provision stipulates that map and navigation apps can collect necessary personal information such as location data and destination, and instant messaging apps are allowed to obtain the registered user's phone number and account details.[48] However, this provision does not specify the scope of necessary personal information that generative AI products can collect. Although Article 11 of the AI Measures fills this gap and reiterates that generative AI service providers "shall not collect non-essential personal data," the definition of "essential" remains ambiguous and is practically determined by the service providers themselves. Consequently, the legal force of this provision is limited. Furthermore, when generative AI is integrated into and used together with other apps, it may weaken the effectiveness of the data minimization principle in regulating app providers' data-gathering activities. For example, if Ernie Bot were incorporated into an educational app, that app may claim to be generative AI based on its content generation function, thus potentially bypassing Article 5 of the Provisions 2021. Given the tension between advancing research and the development of large AI models and protecting personal information, the minimization principle places constraints on the data accessible to generative AI service providers. However, this cautious approach may inadvertently hinder technological progress. This tension is evident in the AI Measures: Article 3 encourages innovation and development of generative AI while emphasizing prudent supervision, while Article 4 mandates that generative AI services must not infringe upon personal information rights and interests. This juxtaposition highlights the conflict between public values and private interests. Resolving this dilemma—specifically, determining the scope of "necessary personal information" in generative AI—requires further thoughtful consideration.

Secondly, the principle of reasonable purpose may not apply to data processing activities of generative AI. According to Article 6(1) of the PIPL, collecting personal data shall have specific, explicit, and legitimate purposes, and data cannot be processed in a manner that is incompatible with those purposes. This provision is comprised of two critical aspects: (1) clarity and specificity; and (2) use restriction.

The first aspect is a procedural restriction that requires the processors to carefully plan data collection activities and inform users of the intended purposes of such activities. In principle, the processors should not arbitrarily change the purposes. However, considering that big data can reveal hidden or unpredictable inferences and correlations from vast datasets, some legislation allows the processors to modify the initial purposes, albeit with certain restrictions. For instance, Article 5.1(b) of the EU's GDPR stipulates that further processing for achieving purposes in the public interest, scientific or historical research purposes, or statistical purposes shall not be considered to be incompatible with the initial purpose. The Chinese law takes an even more lenient approach. According to Article 23 of the PIPL, the processors shall change the initial purposes if they obtain consent from the individual anew. The second aspect represents a substantive constraint on data processing purposes, including adopting methods that minimize the impact on individual

[47] See Provisions on Administration of Algorithmic Recommendation in Internet Information Services 2021. https://www.gov.cn/zhengce/zhengceku/2022-01/04/content_5666429.htm.

[48] See Provisions on the Scope of Necessary Personal Information for Common Types of Mobile Internet Applications. https://www.gov.cn/zhengce/zhengceku/2021-03/23/content_5595088.htm.

rights and interests and prohibiting differential treatment based on data.

It should be noted that neither aspect addresses what constitutes a "reasonable" purpose of data processing. For instance, are for-profit marketing and personalized recommendations considered reasonable purposes? Currently, Chinese scholars generally hold that legitimate processing purposes encompass both public interest and private interest. In general, the following issues pertain to private interest: fulfillment of contracts, protection of vital interests of the individual or other natural persons, internal management, and provision of safe and stable products and services (Liu, 2021). When it comes to generative AI, the purposes of processing are usually providing users with a better dialog experience or helping models improve performance. From the perspective of developing generative AI technology and industry, these terms may be considered reasonable purposes. However, from the perspective of users who contribute personal data, they receive little benefit but face potential data leakage risks. In this circumstance, it is questionable whether generative AI's purposes of processing are reasonable.

Thirdly, the inapplicability of information obligations in the context of generative AI is a concern. Trustworthy AI is a widely accepted standard for AI governance. To promote AI transparency, it requires data and systems of an AI model to be traceable, explainable, and communicable (EuropeanCommission, 2019). Transparency is not a novel requirement for generative AI, as personal data protection legislation often mandates that the processors account for and explain their data processing activities. For instance, Article 13 of the GDPR requires the controllers to provide specific information to ensure fairness and transparency in data processing. Similarly, Article 17 of the PIPL obliges the processors to inform users of critical aspects of data processing, including purpose, manner, and types of personal data. These requirements for explainability naturally extend to generative AI products. However, information obligations are rarely fulfilled, let alone achieving the purpose of the "informed consent" doctrine.

On the one hand, most users lack expertise in computers and data analytics. The complexity of data processing prevents users from genuinely comprehending lengthy and detailed notification instructions, making it difficult for users to make an informed decision (Mantelero, 2014). On the other hand, critical information about generative AI models, such as model structure, algorithms, data processing workflows, and business models, is usually the core trade secret of generative AI (Wachter & Mittelstadt, 2019). Consequently, enterprises may hesitate to explain these matters in detail. Additionally, since enterprises are often unaware of what specific information algorithms might extract from datasets, or what adjustments algorithms might make to process or utilize data (Edwards & Veale, 2017), they may use vague terms to fulfill the information obligations. In summary, while the requirements of processors' information obligations within the existing personal data protection framework and AI transparency under the Trustworthy AI framework formally respect users' right to be informed, they fall short of helping users fully comprehend the substantive data risks and make rational decisions based on a true understanding of those risks.

4. Conclusion and suggestions

While the emergence of generative AI models like Sora and ChatGPT is opening the prelude to the era of general AI, it also brings forth new risks to privacy and personal data rights and interests. This article, considering OpenAI's public information and the working principles of generative AI, fully analyzes the privacy and personal data risks associated with generative AI and studies the shortcomings of the personal data protection system in Chinese law.

The personal data risks posed by generative AI mainly include privacy risks, data leakage risks, and risks to personal data rights and interests. These risks stem from a multitude of factors. First, generative AI technology, while still evolving, remains immature, leading to data leakage risks throughout the stages of data processing, transmission, storage, and application. Second, generative AI not only reshapes the economic structure, human modes of production, and lifestyle but also introduces new privacy challenges. For example, the unrestricted access of smart home devices to users' data subjects individuals to 24-h surveillance. Third, data processing methods in the era of big data and AI differ dramatically from previous practices, rendering existing privacy and personal data protection rules inadequate in addressing emerging issues effectively.

Currently, China's legal system and governance framework are inadequate in addressing the privacy risks posed by generative AI. To start with, generative AI processes vast amounts of publicly available personal information uniquely and implicitly, posing challenges in effectively applying existing rules to protect individuals' privacy rights. Also, despite the high risk of data leakage posed by generative AI systems, doubts persist regarding whether China's law enforcement agencies will impose substantial fines on data breaches. It remains uncertain whether privacy and personal data risks could be mitigated through public law. In addition, existing legislation, law enforcement practices, and judicial practices in China do not provide additional protection for vulnerable parties in data leakage situations. The CCPRC lacks clear rules for damages that would allow victims of a data leakage accident to bring a personal data tort lawsuit, indicating that addressing risks through the private law approach is also difficult. Moreover, codes of conduct for data processing face applicability problems when applied to generative AI, making it hard to protect individuals' data rights and interests.

In pursuit of the goal of "AI serving humanity," many countries are exploring the most effective ways to manage AI-triggered risks and achieve trustworthy AI. According to Article 3 of the Ethical Norms for New Generation Artificial Intelligence,[49] the trustworthiness of AI in China is conceptualized as a multifaceted value system that includes the enhancement of human well-being, the promotion of fairness and justice, the protection of privacy and security, and the assurance of control and reliability. Achieving these objectives necessitates collaborative efforts from technological, legal, and ethical aspects. Meanwhile, relevant measures should align

[49] See Ethical Norms for New Generation Artificial Intelligence. https://cset.georgetown.edu/publication/ethical-norms-for-new-generation-artificial-intelligence-released/.

with the country's political objectives, economic development, and social reform requirements while balancing the interests of all stakeholders, particularly the interests of individuals, the public, and corporations. Individual interests encompass privacy interests, data interests, and other related personal property interests. Public interests include technological progress, social order, and national security. Corporate interests involve the value of data assets and the profits generated for enterprises through the development of generative AI. The AI Measures integrates China's AI ethical norms with the objectives of digital economy governance. The first article explicitly states that generative AI governance must simultaneously protect the legitimate rights and interests of citizens, legal persons, and other organizations, while also safeguarding national security and the public interest. This article, based on the analysis of privacy and personal data risk governance of generative AI from the perspective of China, proposes suggestions for enhancing privacy and personal data protection regulation in the era of generative AI. Given that many countries face similar challenges and share principles, frameworks, and rules to address these issues, the following suggestions also provide important insights into enhancing their national legislation on personal data protection and AI.

4.1. Data security measures

4.1.1. Technical improvement

From a technical perspective, breakthroughs in de-identification, anti-re-identification, anonymization, encrypted storage, and circulation can effectively mitigate the risk of data leakage (Villegas-Ch & García-Ortiz, 2023). While generative AI inevitably encroaches upon personal privacy by capturing users' daily activities and conversations, adopting measures such as data non-disclosure or de-identification can help mitigate risks. This approach aims to strike a balance between protecting individual privacy and meeting enterprises' data utilization needs. Additionally, efforts are being made to address the security risks arising from data flows between multiple enterprises or servers in the context of joint development or commercialization of generative AI. For instance, Article 2(20a) of the EU's Data Act, which aims to create standards and practices for shared or jointly processed data, proposes to establish a common European data space to foster new products, services, scientific research, and civil initiatives.[50] Similarly, the China Academy of Information and Communication Research released the Trusted Industrial Data Matrix 1.0 in 2022, advocating for constructing novel infrastructure specifically designed for trusted, secure, and shared industrial data.[51] The trusted data matrix, as a comprehensive framework for data processing, circulation, and transactions, extends beyond network hardware. It also introduces advanced data security technologies, including multi-party secure computing, trustworthy execution environments, and federated learning to safeguard future data circulation and transactions.

4.1.2. Institutional improvement

Establishing a safeguarding system for data security that encompasses both preventive measures and post-violation punishment is crucial. The key to preventive measures is to build a categorized and graded protection system for data. Currently, different countries and even regions within the same country have their standards for data importance, sensitivity, and risk. However, there are no consistent global standards for a categorized and graded protection system for data. Taking sensitive data as an example, in China, sensitive data refers to personal data that, once leaked or illegally used, may easily cause severe harm to personal or property security, as well as the personal information of minors under the age of 14. The GDPR identifies personal data related to race, ethnicity, religious beliefs, genetic information, and sex-related data as sensitive data. The US Executive Order No.14110 broadly includes personal data on identity, location, habits, and desires as sensitive data. The absence of a well-defined legal framework for implementing a categorized and graded data protection system compels companies to create their standards and best practices. This underscores the necessity for additional legislation. In terms of punishment mechanisms, clear and stringent legal liabilities, along with strengthened legal supervision, effectively encourage enterprises to meet their data security obligations. Amid the AI industry race, certain countries may loosen AI regulations in pursuit of technical innovation, thereby exposing their citizens to privacy and personal data risks and potentially disrupting social order. Consequently, it becomes essential for countries to refine legal responsibilities based on the actual development and application of generative AI. This includes distinguishing between model developers and service providers and specifying the types and circumstances of violations.

4.2. Legal remedies for privacy and personal data infringements

While China has clarified individuals' data rights and the obligations of data processors during personal data processing, the lack of clarity in infringement remedies may directly impact the realization of these rights and undermine the effectiveness of personal data protection rules. To address this gap, future theories, and legislation should prioritize improving rules related to privacy and personal data infringement remedies. First, the remedies for personal data rights and interests should be clarified. Although personal data protection laws grant users rights such as the right to be informed, the right to access, and the right to obtain a copy, the practical exercise of these rights faces challenges. Legislation should provide clearer guidelines for overcoming obstacles when exercising their right to personal information. Second, the public interest litigation system for personal data infringement should be further developed. Data infringement often impacts multiple individuals simultaneously, with collective harm extending to social and public interests.

[50] Regulation (EU) 2023/2854 of the European Parliament and of the Council of 13 December 2023 on harmonised rules on fair access to and use of data and amending Regulation (EU) 2017/2394 and Directive (EU) 2020/1828 (Data Act). https://eur-lex.europa.eu/eli/reg/2023/2854.

[51] Trusted Industrial Data Matrix 1.0. http://www.caict.ac.cn/kxyj/qwfb/ztbg/202201/P020220125561909082218.pdf.

Future legal frameworks should prioritize users' interests by enhancing the public interest litigation system for personal data infringement, ensuring effective regulation of illegal data processing, and providing remedies for data leakage.

4.3. Updates on the modes of privacy and personal data protection

The existing personal data protection framework assumes that users are capable of fully comprehending the information provided by the processors and that processors shall maintain consistent purposes and methods of data processing. However, these two assumptions are not convincing in the context of generative AI. As a consequence, adjustments to the principle of "informed consent" are necessary. This article suggests a data assessment mechanism between processors and users without changing the relevant present rules.

In this mechanism, the data assessment agency acts as an intermediary and works on behalf of users. The mandates of the agency include evaluating and reviewing data processing activities, issuing reports on the legality and rationality of these activities, and maintaining records. In other words, the development and processing activities of generative AI should be submitted to the agency. If the purpose of processing is legitimate and reasonable and the processing method will not jeopardize user rights, the agency will issue a positive evaluation report. Otherwise, a negative evaluation report will be provided. Correspondingly, the assessment results become part of the user agreement, where users authorize processors to process their personal data based on their trust in the agency.

The data assessment mechanism evolves from the data protection officer system designated in the GDPR, which meets the auditable requirements in the Trustworthy AI framework. To date, personal data protection legislation worldwide essentially requires processors to designate personnel as personal data protection officers. Their duties include considering the risks of data processing, supervising processing activities, providing advice on data protection impact, and cooperating with supervisory authorities. Considering the requirements of data security, transparency, explainability, and auditability advocated by Trustworthy AI, it is advisable to establish an assessment agency independent of the controller or processor to fulfill its mandates (EuropeanCommission, 2019).

4.4. International cooperation on AI regulation

The ongoing digitalization and AI revolution have ushered in a new era of global competition. As economies vie for AI leadership, these countries may relax their regulations of AI, potentially leading to detrimental and irreversible consequences. In previous waves of technological advancement, countries prioritizing unilateral technological development and economic growth often overlooked the irreversible harm caused by data and algorithms to users and society at large. For instance, a significant amount of personal data is still accessible on the "dark web." Therefore, sacrificing governance for technological supremacy in the race for AI is not advisable. Among different development strategies and policies for AI, the GDPR, the EU Data Act, and the Artificial Intelligence Act of the EU have global demonstrative effects, which are inseparable from the EU's endeavors to lead in AI regulation. In terms of the Trustworthy AI objective, enhanced technical regulation for AI is essential, even if achieving this objective is still in its infancy. The globalization of AI products and cross-border flows of data further necessitate more unified technical standards, regulatory concepts, and frameworks, calling for increased international conventions on AI development and regulation.

Declaration of competing interest

We have no known conflict of interest to disclose.

CRediT authorship contribution statement

Xiongbiao Ye: Writing – review & editing, Writing – original draft, Supervision. **Yuhong Yan:** Writing – review & editing, Writing – original draft. **Jia Li:** Writing – review & editing, Formal analysis. **Bo Jiang:** Visualization, Supervision, Formal analysis.

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